Verifying the Robustness of Machine Learning based Intrusion Detection Against Adversarial Perturbation

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***Abstract*—Neural networks (NNs) have been extensively adapted to various security tasks, such as spam detection, phishing, and intrusion detection. Particularly in IDS, NNs face significant vulnerabilities to adversarial attacks, where the adversary attempts to exploit the fragilities within machine-learning (ML) models. This study introduces a novel approach using interval-bound propagation (IBP) to formally verify and enhance the resilience of both shallow and deep NNs. We also investigated the effectiveness of various activation functions using benchmark IDS datasets. Our findings show that ReLu and Leaky-ReLu functions enhance resistance in shallow networks, whereas Tanh functions perform better in deep networks. We provide certified accuracies for models subjected to various input perturbations ranging from 0.0001 to 0.9, marking a significant advancement in verifying the security of NNs.**

***Index Terms*—Cybersecurity, Neural Network Security, Adversarial Robustness, Intrusion Detection System, Formal Verification, Robustness, and Deep Learning**

I. Introduction

Artificial intelligence (AI) has been progressively integrated into enterprise cybersecurity intrusion detection systems (IDS) to keep up with the complexity and volume of modern cyber threats [1]–[6]. Deep neural networks (DNNs) are currently widely employed in IDS implementations to perform significant tasks such as spam filtering, malware detection, network anomaly recognition [7], and attack classification [8], [9]. Nevertheless, the technical complexity and black-box features associated with DNNs result in reasonable concerns regarding reliability, security, confidence, and safety during their implementation [10], [11].

First, the absence of transparency and explicability in models of complex NNs results in uncertainties regarding their failure, thereby complicating the prediction of such occurrences [12]. The black-box nature of DNNs conceals the interactions between inputs and outputs, thereby providing a limited understanding of the underlying logic that drives predictions or classifications [13]. The absence of visibility in the logic and boundaries of decisions within a model hinders experts from employing dependable techniques to validate accuracy or define modes of failure [14], [15]. The demand for explainability is a leading priority in high-assurance defense techniques, which offers significant challenges for verification and auditing processes [16] and also plays a pivotal role in different domains [17] such as health care systems, cyber-physical methods, and so on.

Likewise, scholars have demonstrated that DNNs are exceptionally vulnerable to adversarial examples, which consist of intentionally corrupted inputs that result in high-confidence misclassifications, but appear unmodified to humans [18]. Adversaries can employ various adversarial attacks to degrade the performance of the model or any kind of error in a classification by taking advantage of the blind areas attending in high-dimensional ML models [19]. In this case, security concerns increase from these adversarial attacks, which affect the robustness of IDS approaches when adversaries attempt to bypass correct detection or apply backdoors during model training [20]. Besides, withstanding adversarial attack methods is a challenging task due to the lack of knowledge concerning DNN decision boundaries [21]. Recently, several research studies have drawn attention to the increasing threats and vulnerabilities that adversarial learning introduces into domains that are essential to user safety, such as anomaly monitoring systems and malware detection [20].

Consequently, demanding verification techniques that can provide formal guarantees regarding the performance of machine learning models employed in IDS infrastructure has become essential [22]. This research presents an innovative verification method that quantifies the resilience of shallow and deep NNs against adversarial attacks by employing interval-bound propagation (IBP) [23]. This method enables the formal verification of NN robustness by estimating lower and upper bounds on network activations and outputs.

A. Research Questions (RQ)

The main objective of this research is to assess the resilience of intrusion detection systems (IDS) based on machine learning against deliberate attacks by utilizing interval-bound propagation (IBP) as a formal verification technique. Specifically, this study aims to:

RQ1) What is the efficacy of interval-bound propagation (IBP) in improving the resilience of machine learning models employed in IDS against adversarial attacks?

RQ2) Which NN designs and activation functions exhibit the highest resistance to adversarial perturbations?

RQ3) How well do the proposed approaches perform when tested on realistic intrusion detection datasets?

B. Contribution

In short, this study has the following novel contributions:

- This study employed shallow and deep neural networks to provide IBP for AI robustness verification in an IDS. IBP was selected based on its demonstrated effectiveness in prior research [23], which showed its ability to reliably and efficiently evaluate the robustness of NN models against adversarial attacks. This study leverages IBP to provide a formal verification of model robustness in the context of IDS.
- The popular benchmark IDS datasets, CICIDS2017 and UNSW-NB15, were employed to assess the robustness of the ReLu, Leaky-ReLu, and Tanh activation functions in NN models. Although the relevance and applicability of these datasets to real-world intrusion detection settings have been previously established [24]–[26], our study leverages them to evaluate the robustness of models under adversarial perturbations using interval-bound propagation (IBP).
- These findings indicate that models incorporating the ReLu and Leaky-ReLu activation functions are more resistant to adversarial attacks in shallow networks. Conversely, deep networks perform better when the Tanh activation functions are applied.
- An important step forward in the validation of AI security in NNs for IDS is the provision of certified accuracy for models subject to a range of input perturbations from 0.0001 to 0.9.

C. Organization

We outline the rest of our paper as follows: In Section II, we provide the background and related works recently applied to IDS in the presence of machine/deep learning, and then we investigate recent approaches that consider IBP as a verification method. In Section III, we provide our methodology, threat model, and preliminaries on datasets such as CICIDS2017 and UNSW-NB15. Moreover, in this section, we provide information regarding network architectures that we refer to as shallow networks when we have only five hidden layers and a deep network when we have 10 hidden layers. Section IV presents the settings and experimental results, followed by a discussion of the certified accuracy. Finally, in Section V, we conclude our study and provide promising future research.

II. Background and Related Works

In this section, we present an overview of the IDS and the IBP verification method, along with the related works.

A. Intrusion Detection System

IDS are essential for identifying and preventing potential threats and abnormalities in network traffic and systems [27]. Owing to the growing complexity of attacks, IDS systems have extensively incorporated machine learning and deep learning models for the purposes of detection, classification, and analysis [28]. Nevertheless, some AI models have shown susceptibility to adversarial attacks that may bypass detection or impact outcomes [29]. Therefore, the study of model robustness and security has become the highest priority. Recent studies have emphasized the use of formal verification techniques such as reachability analysis [30], SMT solvers like Reluplex [22], and interval bound propagation (IBP) [23] to assess and improve the resilience of NNs against adversarial attacks. Reachability analysis involves computing the set of all possible outputs given a set of inputs, ensuring the NNs robustness across various scenarios. Reluplex, an efficient SMT solver, has been particularly effective in verifying the safety of DNNs by solving linear programming problems.

B. Formal Verification

To prove that deep NNs are robust against adversarial attacks, we considered the IBP method, which is one of the recent formal verification techniques that applies interval arithmetic [31]. The IBP method is implemented layer-by-layer, propagating input sets across the NN network to estimate lower and upper bounds on activations and final outputs [22]. Figure 1 illustrates the layer-wise IBP bound propagation, where lower and upper values for parameters such as weights and biases are considered across the neurons of the same layer. For a fully connected layer with ReLu activation, the IBP approach is outlined in the pseudocode provided in Algorithm 1. In this algorithm, the variables x_{min} and x_{max} are related to the lower and upper bounds of the input data features, rather than simply the input data itself. The W_{min} and W_{max} refer to NN minimum and maximum weights values, the b_{min} and b_{max} to biases, the z_{min} and z_{max} for pre-activation values, and a_{min} and a_{max} related to minimum and maximum values of post-activation such as ReLu.

Algorithm 1: IBP Formal Verification Method

Input: $x_{\text{min}}, x_{\text{max}}, W_{\text{min}}, W_{\text{max}}, b_{\text{min}}, b_{\text{max}}$
Output: $a_{\text{min}}, a_{\text{max}}$
Compute pre-activation bounds:
$z_{\text{min}} \leftarrow (W_{\text{min}} \cdot x_{\text{min}}) + (W_{\text{max}} \cdot x_{\text{max}}) + b_{\text{min}}$
$z_{\text{max}} \leftarrow (W_{\text{min}} \cdot x_{\text{max}}) + (W_{\text{max}} \cdot x_{\text{min}}) + b_{\text{max}}$
Compute post-activation bounds:
$a_{\text{min}} \leftarrow \max(0, z_{\text{min}})$
$a_{\text{max}} \leftarrow \max(0, z_{\text{max}})$
return $a_{\text{min}}, a_{\text{max}}$

C. Research Gaps (RG)

While previous studies, including [23], have demonstrated the effectiveness of IBP in training verifiably robust models, there remains a gap in the literature regarding the application of these techniques specifically to IDS. Our study addresses this gap by:

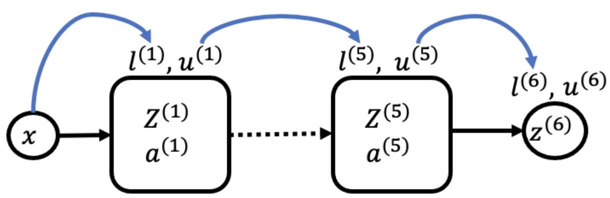

Fig. 1: Formal Verification Illustration: IBP Methodology for Layer-by-Layer Estimation of Lower and Upper Bounds in a Five-Layer Network.

RG1) Utilizing the IBP technique to evaluate the resilience of IDS models to adversarial attacks, a topic that has not been fully explored in previous studies.

RG2) Evaluating the efficiency of various NN models and activation functions in IDS, offering valuable insights into their relative robustness.

RG3) Employ real IDS datasets (CICIDS2017 and UNSW-NB15) to ensure the practical significance and relevance of the findings.

RG4) Providing precise certified accuracies for various input perturbations and improving the understanding of model behavior in adversarial settings.

III. Methodology

The primary goal of this research is to evaluate the robustness of machine learning-based IDS against adversarial attacks [32]. This is achieved by employing interval-bound propagation (IBP) as a formal verification method. The process begins with the collection and preprocessing of Pcap files containing the required network traffic data, utilizing NFStream [33] to extract relevant features. In this preprocessing stage, the data is normalized to ensure its integrity and suitability for further analysis. During this stage of preprocessing, the data were normalized to guarantee their integrity and appropriateness for further analysis. Subsequently, the extracted features from the NFstream are utilized to train the NN models, taking into account various structures, including shallow and deep networks. The robustness of these models is investigated by training them with various activation functions, such as ReLu, Leaky-ReLu, and Tanh, by considering the IBP approach, showing which NN model is robust. To mitigate concerns related to class imbalance, strategies such as the implementation of class-weight computations [34] are employed.

The key focus of this study is the use of interval-bound propagation (IBP) as a formal verification technique, which allows for the accurate estimation of the lower and upper bounds of the model's activations and outputs. This ensures resilience against adversarial attacks by providing certified accuracy at various perturbation levels. In fact, constraints are determined based on the characteristics and limitations of the input data. Interval boundaries are propagated throughout the layers of the NN using IBP, which offers formal guarantees of the model's performance and ensures resilience against adversarial attacks, although it does not offer complete and absolute assurance in all circumstances and for all types of adversarial samples. After deploying IBP, this study focused on assessing the certified accuracy of the models. This assessment entails comparing the predicted outputs of the model with the interval boundaries calculated using IBP. The robustness of the models against adversarial attacks is evaluated by manipulating the input constraints and epsilon values. The term "input constraints" refers to the limitations or restrictions made on input data throughout the propagation of interval boundaries through the layers of an NN, whereas epsilon values in IBP influence the amount of perturbation that may be imposed on the input data.

A. Threat Model

The threat model implies that attackers try to undermine the security and reliability of neural-based IDS through the development of adversarial examples that exploit the system's fragility by providing perturbations. Adversaries deliberately generate instances within the epsilon constraints of distortion to mitigate the robustness and accuracy guarantees offered by the IBP-verified approach. The IBP derives predictive certainty through the propagation of input intervals throughout network layers; hence, threat targets cause the IBP's lower and upper prediction limits to differ from the models' actual outputs for perturbed inputs. The ability of IBP to verify its efficacy under adversarial distortion scales presents a potential threat, as it compromises the reliability of certifications regarding the security of intrusion detection models and the ability to withstand evasion attacks via formal verification. An in-depth investigation of adversarial threats has been carried out on standardized datasets using shallow and deep network architectures, each applying a unique activation function. This evaluation is intended to validate the robustness of NNs utilized by IBP in critical infrastructure safety applications, like intrusion detection.

B. Datasets

To assess the robustness of the models in the field of intrusion detection, we studied wide real-world datasets for IDS that included both malevolent and benign examples. It is important to emphasize the need for high-quality datasets to achieve a high level of training accuracy. Our study included training multi-perception shallow and deep networks using the CIC [35] and UNSW [36] datasets. Moreover, these datasets are commonly known and have recently been utilized for several cybersecurity objectives. The CIC datasets provide genuine network traffic data, including both benign and malicious instances, to enable experts to build and assess efficient IDS. Moreover, UNSW (UNSW-NB15) offers a wide range of network traffic data, including many types of attacks, such as DoS and worms, similar to the CIC datasets.

In the next section, we elaborate on the specific subsets of the dataset that were leveraged in this study.

*1) **CICIDS2017**:* This dataset [35] was developed by academics and industry experts to serve as a comprehensive benchmark for evaluating IDS in cyber-physical systems. It contains real-world network traffic, encompassing both benign and malicious activities. The data incorporates common contemporary threats like denial-of-service (DoS), distributed DoS (DDoS), and reconnaissance attacks to demonstrate various attack types. Rigorous preprocessing procedures were followed to ensure quality and utility, including feature extraction, normalization, and anonymization. In addition, carefully selected and accurately labeled samples were collected to facilitate the development and assessment of supervised intrusion detection models. The primary motivation behind creating CICIDS2017 was to meet the need for standardized datasets in cybersecurity. This valuable resource enables researchers to assess and improve the efficacy of IDS.

*2) **UNSW**:* This dataset [36] was generated from the Cyber Range Lab using the IXIA PerfectStorm tool. It incorporates a hybrid of real, up-to-date normal traffic and synthetically generated contemporary attack patterns. Two servers handled regular network activities, while a third server launched various attacks like denial of service (DoS), worms, and backdoors. The traffic was captured and analyzed, leveraging technologies including Argus and Bro-IDS to extract 49 features categorized as flow-based, content-based, and time-based. This rigorous process resulted in a substantial dataset encompassing both benign and malicious network behaviors, offering valuable insights for intrusion detection research.

C. Neural Network Architecture

In our experiments, we consider two main multilayer perceptron NNs. In this study, we refer to one as a shallow network when it has 5 hidden layers and the other as a deep network when it has 10 hidden layers. We considered twelve scenarios to understand the circumstances under which IDSs and NNs are robust against adversarial attacks. Moreover, to understand the robustness of the networks, we considered different activation functions, such as ReLu, Leaky-ReLu, and Tanh. Additionally, four scenarios were trained using the CIC dataset, and four scenarios were trained using the UNSW dataset. The architectural details of the scenarios are as follows in Table I.

- **Shallow Layer Network:** The configuration of the shallow network consists of five hidden layers, with the number of neurons considered in each layer being **[512, 256, 128, 64, 8]**. The pipeline of the shallow network is shown in Figure 2. For the shallow network, we considered four scenarios, as listed in Table I. In CIC5-ReLu, the dataset is CIC and the activation function is ReLu. In CIC5-Leaky-ReLu, the dataset is CIC, but the activation function is Leaky-ReLU. The same procedure is applied to the UNSW datasets UNSW5-ReLu and UNSW5-Leaky-ReLu, and also to CIC5-Tanh and UNSW5-Tanh.
- **Deep Layer Network:** The configuration of the deep network consists of 10 hidden layers, with the number of neurons considered in each layer being **[1024, 512, 512, 256, 256, 128, 128, 64, 32, 16]**. For the deep network, we considered four scenarios similar to the shallow network, as listed in Table I. In CIC10-ReLu, the dataset is CIC, and the activation function is ReLu. In the CIC10-Leaky-ReLu, the dataset is CIC, but the activation function is a Leaky-ReLu. The same procedure is applied to the UNSW datasets UNSW10-ReLu and UNSW10-Leaky-ReLu, and the same strategy is also applied to CIC10-Tanh and UNSW10-Tanh.

TABLE I: Outline of the networks and datasets were considered in this study.

Sce.	**Net.**	**Dataset**	**Act. Func**	**Called**
1	Shallow	CIC	ReLu	CIC5-ReLu
2	Deep	CIC	ReLu	CIC10-ReLu
3	Shallow	CIC	Leaky-ReLu	CIC5-Leaky-ReLu
4	Deep	CIC	Leaky-ReLu	CIC10-Leaky-ReLu
5	Shallow	CIC	Tanh	CIC5-Tanh
6	Deep	CIC	Tanh	CIC10-Tanh
7	Shallow	UNSW	ReLu	UNSW5-ReLu
8	Deep	UNSW	ReLu	UNSW10-ReLu
9	Shallow	UNSW	Leaky-ReLu	UNSW5-Leaky-ReLu
10	Deep	UNSW	Leaky-ReLu	UNSW10-Leaky-ReLu
11	Shallow	UNSW	Tanh	UNSW5-Tanh
12	Deep	UNSW	Tanh	UNSW10-Tanh

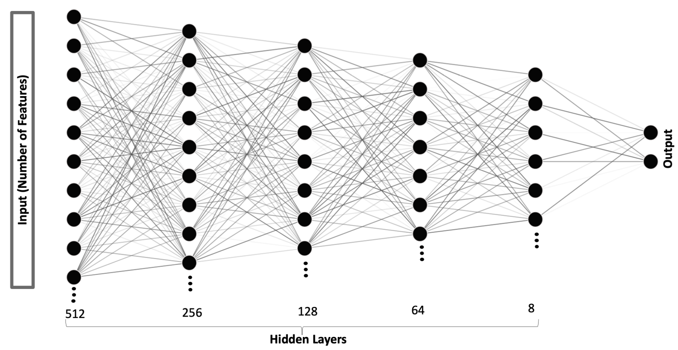

Fig. 2: Pipeline of the Shallow Network.

D. key performance indicators (KPIs) of robustness

Robustness, in the context of this study, refers to the potential of machine learning-based IDS to maintain remarkable performance and accuracy even when exposed to adversarial perturbations. Ensuring the robustness of an IDS is essential for consistently identifying and mitigating security threats across different situations. The KPIs used to assess the level of resilience in this study are as follows:

Certified Accuracy: The percentage of inputs for which the model's predictions are demonstrably reliable within the specified perturbation bounds.

Adversarial Robustness: The capability of the model to withstand adversarial attacks without a significant drop in performance, as measured by the decrease in accuracy when exposed to adversarially perturbed inputs.

Resilience of Various Architectures and Activation Functions: A Comparative Analysis of the Robustness of Various NN Architectures (Shallow and Deep) and Activation Functions (ReLu, Leaky-ReLu, Tanh) against adversarial attacks.

E. Configuration of Hardware and Software

The experimental setup consisted of training a NN model on two datasets, CIC and UNSW, using Python and PyTorch, respectively. The CIC dataset consists of 65 columns and 1,048,575 samples, of which 1,014,818 are labeled zero (pristine) and 33,757 are labeled one. The UNSW dataset comprises 76 columns and 1,048,574 samples; 95,939 features were assigned a value of one, while 952,635 features were assigned a value of zero. The training process for these datasets included 40 epochs and employed the Adam optimizer, which had a batch size of 32 and a learning rate (Lr) of 1e-4. The Adam optimizer was chosen because of its adaptive learning rate capabilities and efficiency in handling sparse gradients, which are common in high-dimensional data such as network traffic features. Although SGD with Momentum or Nesterov Accelerated Gradient (NAG) could also be effective, Adam provides faster convergence and better performance in practice for this specific application. To rectify class imbalances, the class weights are calculated. The outcomes of the training and testing procedures are stored for further examination on an NVIDIA GPU that supports CUDA. Based on the twelve scenarios in Table I, ReLu activation functions were considered in four scenarios, and leaky-reLu and Tanh activation functions were employed in the remaining scenarios. During the evaluation process, IBP is employed to determine the certified accuracy and evaluate the robustness of the model. By propagating the IBP bounds throughout the NN model, the lower and upper bounds for estimations are computed, which indicate the model's confidence and trustworthiness in real-world scenarios. The datasets of both CIC and UNSW are split as follows: 60% of the data is dedicated to training, 20% to validation, and 20% to testing. The programmed code is available in [37].

IV. Evaluation and Discussion

This section provides the experimental results regarding the scenarios in the absence of attacks, assesses the robustness of IDS models in different perturbations (referred to as epsilon), and reports the accuracy of models trained with CIC and UNSW in a shallow and deep network.

A. Networks Training Process in the Absence of Attacks

In Table II, the IDS model demonstrates high-accuracy metrics in training, validation, and testing for almost all types of NNs, even in the absence of adversarial attacks. When employing the CIC datasets with 5-layer shallow networks and the Tanh or ReLu activation functions, the testing accuracy exceeded 99%. All UNSW dataset models exhibited remarkable testing performance, achieving an accuracy of approximately 99-100% accuracy. Only the more complex 10-layer CIC models with ReLu or LeakyReLu activation achieved a lower testing accuracy of, 96-97%. Owing to model complexity, NNs are more likely to overfit and are less able to generalize to new data, resulting in a slight decrease in accuracy. The 10-layer models' increased number of layers and parameters may allow slight perturbations that affect the output, leading to decreased testing accuracy. Given the complexity of the dataset and the strength of the models, we believe that the accuracy is feasible.

TABLE II: Performance of IDS models in the absence of attacks.

Scenarios	Train Acc.	Validation Acc.	Testing Acc.
CIC5-ReLu	0.9964	0.9965	0.9963
CIC10-ReLu	0.9837	0.9635	0.9640
CIC5-Leaky-ReLu	0.9663	0.9660	0.9667
CIC10-Leaky-ReLu	0.9653	0.9651	0.9657
CIC5-Tanh	0.9916	0.9912	0.9913
CIC10-Tanh	0.9932	0.9934	0.9932
UNSW5-ReLu	0.9999	0.9999	0.9999
UNSW10-ReLu	0.9999	0.9999	0.9999
UNSW5-Leaky-ReLu	0.9998	0.9997	0.9998
UNSW10-Leaky-Relu	0.9993	0.9992	0.9993
UNSW5-Tanh	0.9998	0.9999	0.9999
UNSW10-Tanh	0.9996	0.9995	0.9963

B. Shallow Networks Certified Accuracy

As shown by the certified accuracy in Table III, the shallow networks exhibit reasonable robustness across a range of epsilon perturbation values. All shallow networks, excluding the CIC5-Tanh and UNSW5-Tanh models at the 0.1 epsilon level, achieved a certified accuracy of over 70%, despite the maximum distortion level. This finding implies that the remarkable robustness to input perturbations is caused by the ReLu/LeakyReLu activation functions and shallow structure. In particular, in comparison to the Tanh function, the ReLu and LeakyReLu activation functions generated significantly more robust results. Significant variations exist in the formulation of these activations: Tanh bounds outputs between -1 and 1, whereas ReLu and LeakyReLu are capable of generating unbounded values. Unlike Tanh, which compresses variances via its nonlinear formulation, the dynamic ranges of ReLu/LeakyReLu likely assist in the accurate propagation of such behaviors. The UNSW5-ReLu network demonstrated remarkable robustness by maintaining a verified accuracy of over 90% at an epsilon value of 0.1. Although the CIC5-Tanh and UNSW5-Tanh scenarios showed drastically decreased certified accuracies at an epsilon value of 0.1, they nevertheless demonstrated satisfactory performance at lower epsilon values.

Thus, shallow networks often perform well as reliable IDS, particularly when faced with distorted adversarial inputs. The findings confirm that ReLu and LeakyReLu activations provide substantial improvements in adversarial robustness over Tanh in shallow network IDS. In this case, adversarial training methods may improve resilience, particularly for Tanh activation models with larger epsilon values. Overall, the shallow network results affirmed the capability to build verification-validated AI security.

TABLE III: Certified Accuracy in Shallow Network

Scenario	Epsilon	Certified Accuracy
CIC5-ReLu	0.0001	0.9963
	0.001	0.9961
	0.01	0.9919
	0.1	0.5649
UNSW5-ReLu	0.0001	0.9999
	0.001	0.9999
	0.01	0.9999
	0.1	0.9223
CIC5-Leaky-ReLu	0.0001	0.5701
	0.001	0.5700
	0.01	0.5686
	0.1	0.5551
UNSW5-Leaky-ReLu	0.0001	0.7196
	0.001	0.7195
	0.01	0.7188
	0.1	0.7107
CIC5-Tanh	0.0001	0.9205
	0.001	0.9142
	0.01	0.8086
	0.1	**0.0205**
UNSW5-Tanh	0.0001	0.9783
	0.001	0.9781
	0.01	0.9205
	0.1	**0.0915**

C. Deep Networks Certified Accuracy

Table IV on Certified Accuracy in Deep Networks reveals considerable vulnerability in deepest IDS networks as the epsilon distortion levels increase. Ten-layer models often exhibit significant decreases in certified accuracy when faced with input perturbations of 0.001 epsilon, unlike the shallow models. The UNSW5-Tanh structure is remarkable for its strength, maintaining certified accuracy levels of over 90%, even when the epsilon values are as high as 0.1. Pairing a Tanh activation with a deeper architecture results in better and more reliable decision boundaries for adversarial samples from the UNSW dataset. No other deep network structure has achieved a certified accuracy of over 50% when subjected to an epsilon perturbation greater than 0.0001. These variations demonstrate how the dataset complexity and activation functions affect the vulnerability of the model to propagating distortions.

The Tanh activation function, with its symmetric properties and output range between -1 and 1, may be particularly effective for the specific patterns and features present in the UNSW dataset. These specific patterns include varied attack types and sophisticated normal traffic simulations, which benefit from the ability of the tanh function to capture both positive and negative activations, thereby enhancing the decision boundaries in the feature space. A deep network utilizing Tanh activations will cope with the features and distribution of the UNSW dataset, which includes both benign and complex attack types. This alignment allows the deep network to learn and generalize from different inputs, strengthening its accuracy.

TABLE IV: Certified Accuracy in Deep Network

Scenario	Epsilon	Certified Accuracy
CIC5-ReLu	0.0001	0.8547
	0.001	0.0333
	0.01	0.0322
	0.1	0.0321
UNSW5-ReLu	0.0001	0.9999
	0.001	0.5159
	0.01	0.0007
	0.1	0.0000
CIC5-Leaky-ReLu	0.0001	0.3751
	0.001	0.0074
	0.01	0.0023
	0.1	0.0004
UNSW5-Leaky-ReLu	0.0001	0.9490
	0.001	0.0941
	0.01	0.0003
	0.1	0.0000
CIC5-Tanh	0.0001	0.9806
	0.001	0.0339
	0.01	0.0332
	0.1	0.0321
UNSW5-Tanh	0.0001	0.9992
	0.001	0.9985
	0.01	0.9091
	0.1	0.9085

V. Conclusion and Future Works

This paper presents an innovative utilization of IBP to validate the robustness of NN intrusion detection systems against adversarial attacks. Unlike previous studies, such as [32] and [23] , which primarily focused on general machine learning models, our research specifically applies IBP to the context of IDS. We compared the robustness of shallow and deep NN models using different activation functions (ReLu, Leaky-ReLu, and Tanh) on real-world IDS datasets (CICIDS2017 and UNSW-NB15). This approach allows us to provide precise certified accuracies under various perturbation levels, a contribution that has not been extensively covered in previous research. Our findings indicate that shallow networks with ReLu and Leaky-ReLu activations demonstrate significant robustness, whereas deep networks benefit from Tanh activations, especially with the UNSW dataset. Although our approach signifies a significant progression in establishing trustworthy and accountable artificial intelligence to protect critical infrastructure, certain limitations need to be addressed. The stability of deep networks under high perturbations remains a challenge, particularly with respect to Tanh activation functions. Future research should focus on exploring advanced activation functions and defensive distillation techniques to enhance the robustness of deep networks [38]. Additionally, incorporating real-time adaptive mechanisms and further comparisons with emerging verification techniques are crucial for improving the practical applicability and reliability of IDS models in dynamic cybersecurity landscapes.

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